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A paleomagnetic study of the Bilecik Limestone (Jurassic), northwestern Anatolia

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Paleomagnetic analyses of samples collected from a 500 m thick Jurassic section in the Pontides reveal the presence of two components of remanent magnetization: an unstable, low-temperature component which is removed during thermal demagnetization through 220°C and a dominant component which displays consistent directions through 650°. Curie point and IRM studies indicate that goethite is responsible for the low-temperature component whereas both magnetite and hematite contribute to the more stable component. The pole position determined from the stable magnetization is located at 18.8°N, 91.8°E ($\alpha_{95} = 7.7°$, $N = 134$) indicating that the section has undergone more than 90° clockwise rotation since the Late Jurassic. Ancillary geologic evidence, particularly the orientation of Jurassic facies belts is also consistent with a 90° clockwise rotation in this region of northwest Anatolia. The pole suggests that the section may also have migrated slightly northward. Although the age of these movements is currently unknown, it is proposed that they are principally related to the closure of the Neo-Tethys during the Late Cretaceous/Early Tertiary. Some of the rotation may be related to the right lateral movement along the North Anatolian Transform Fault which was initiated in the Miocene.

1. Introduction

The evolution of the Neo-Tethyan realm has provided one of the classical debates for generations of earth scientists. Many models have been proposed to explain the origin and evolution of this complex region extending from the Swiss Alps to the Himalayas. The present-day tectonic elements of Turkey probably were geographically scattered throughout the Tethyan Ocean during much of the Mesozoic, being assembled to produce a single geographic entity only during the early to middle Tertiary [1–3]. The geologic history of Turkey, therefore, is intricately associated with this larger picture, and the Jurassic records the events associated with the early evolution of this region.

Paleomagnetism has been used extensively to determine the motions of large and small crustal blocks (e.g. [4–9]). More recently the technique has been successfully extended to document the timing and amount of crustal deformation in orogenic zones (e.g. [10–12]). Whilst there are problems associated with determining block rotations from limited paleomagnetic data [13], in principle the technique is a valuable tool in unravelling the complicated movements which occur in orogenic belts. Although within the Tethyan realm several such studies have been carried out (e.g. [14–18]), to date, very little paleomagnetic research has been conducted in Turkey [19]. Only three studies of Mesozoic age rocks from Turkey have been reported [20–22]. The paleomagnetic study of Jurassic age shelf carbonates described in this paper was designed to provide information concerning the tectonic history of a small part of northwestern Anatolia and to allow the magnetostratigraphic comparison of these rocks with exist-

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ing Jurassic magnetostratigraphic sections. The results of the stratigraphic study will be published elsewhere.

2. Geologic setting

Turkey can be subdivided into four major tectonic provinces (Fig. 1) (23): (1) the region of Border Folds, (2) the Taurides, (3) the Anatolides, and (4) the Pontides.

The Pontides occupy all of northern Turkey, extending from the Aegean in the west to the Russian border in the east. This study will concentrate on the northwestern portion of this orogenic belt.

The Jurassic of the northwestern Pontides occupies an outcrop belt that extends from the Aegean coast eastward to the town of Nallihan (Fig. 2). At this location, the Jurassic is covered by an extensive blanket of Neogene volcanics. To the north, the Jurassic is buried by Cretaceous and Tertiary units, and to the south Jurassic outcrops are terminated by thrust faults that mark the Pontide-Anatolide boundary in this area.

The Jurassic of this region is characterized by two distinct lithofacies:

(a) Detrital clastics—the Bayırköy Sandstone is a yellow/brown, sparsely fossiliferous, coarse-grained sandstone. In a few localities the basal part of this unit is conglomeratic, containing reworked fragments from the underlying Paleozoic

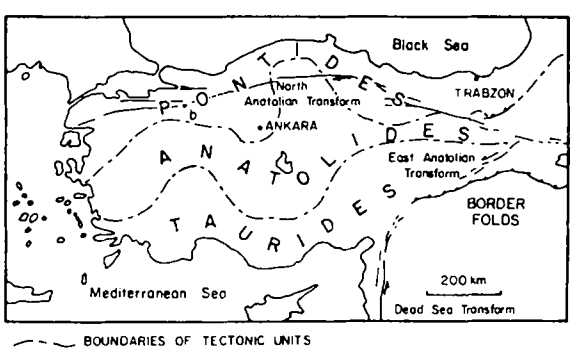


Fig. 1. Major tectonic units of Turkey. *b* = Bilecik sample site. Modified after Ketin [23].

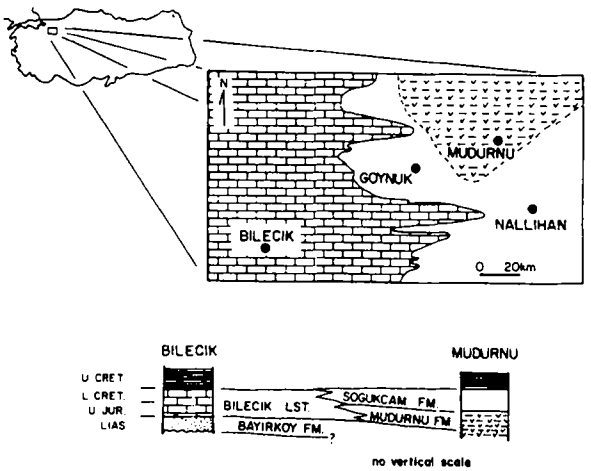


Fig. 2. Late Jurassic lithofacies map of the Mudurnu basin, northwest Anatolia. Modified after Saner [28]. Bayırköy Formation: detrital clastic coarse-grained sandstone; Bilecik Limestone: shelf limestone; Mudurnu Formation: volcaniclastic sandstones and shales; Sogukcam Formation: deep shelf to basinal limestone.

basement. The Bayırköy Sandstone is assigned a Liassic age [24].

(b) Carbonates—the Bilecik Limestone is a buff/grey micrite-peloidal packstone that commonly unconformably overlies the Bayırköy Sandstone, and in some places lies directly on the deformed Paleozoic basement. This unit is characterized by regular, uniform, medium (10–30 cm) to thick (30–100 cm) beds of bioturbated limestones that contain almost no siliciclastic debris (< 1%). Chert nodules and thin (5–10 cm) beds of chert commonly occur within this unit. The age of the Bilecik Limestone is uncertain. Several workers have suggested that it is of Dogger-Malm age [24–26]. Brinkmann [27] suggested that the Bilecik Limestone is principally of Malm age.

The Bilecik Limestone was deposited on a shallow marine shelf that formed part of the margin of the Mudurnu-Goynuk Basin (Fig. 2) [28]. The associated Jurassic facies belts of northwest Turkey today are oriented north-south with the platform/shelf facies to the west and the slope/basin facies to the east. The relationship shown by Saner [28] is a fairly simple shelf-basin transition. Recent unpublished work by geologists of the Turkish Minerals Research and Exploration

Institute (M.T.A.) suggests that the basin margin is more complex with the recognition of a north-south reef trend that passes eastward into a slope/basin facies characterized by carbonate turbidite deposits that contain gravel size bioclastic/boundstone debris. There is no significant amount of detrital clastic debris associated with the limestone suggesting that the platform was either geographically isolated from a major land area, or the associated land mass was topographically mature and/or an arid climate prevailed. The stratigraphic section measured near Bilecik indicates a depositional setting that probably evolved from a deeper water open shelf characterized by low-energy mudstone and peloidal wackestone deposits to a shoaler water shelf setting characterized by higher energy packstone and grainstone deposits. This coarsening upward sequence contains evidence of higher energy conditions in the form of "rip up" clasts, increasing amounts of coated grains and coarse bioclastic debris containing a few boundstone fragments.

3. Paleomagnetic analysis

Sampling. A total of 191 samples were collected from an approximately 500-m-thick section in the Bilecik Limestone located near Deresakarı (40°9′N, 30°8′E) (Fig. 3). Samples were collected at 5-m intervals in the lowermost 250 m of the section, at 1.5-m intervals in the middle 100 m and at 5-m intervals in the uppermost 50 m of the section. At the base of the section and at the top, seven samples were collected from a single bed.

Sample treatment and magnetic mineralogy. The magnetization of the samples was measured using a two-component cryogenic magnetometer at the Institute for Geophysics, University of Texas at Galveston. Two independent estimates of magnetization were obtained for each sample. The vector difference between these estimates and/or the noise level of the magnetometer ($\sim 2 \times 10^{-5}$ A/m) provided a measure of the reliability of the magnetization obtained.

Natural remanent magnetization (NRM) intensities measured are bimodal (Fig. 4a) with maxima

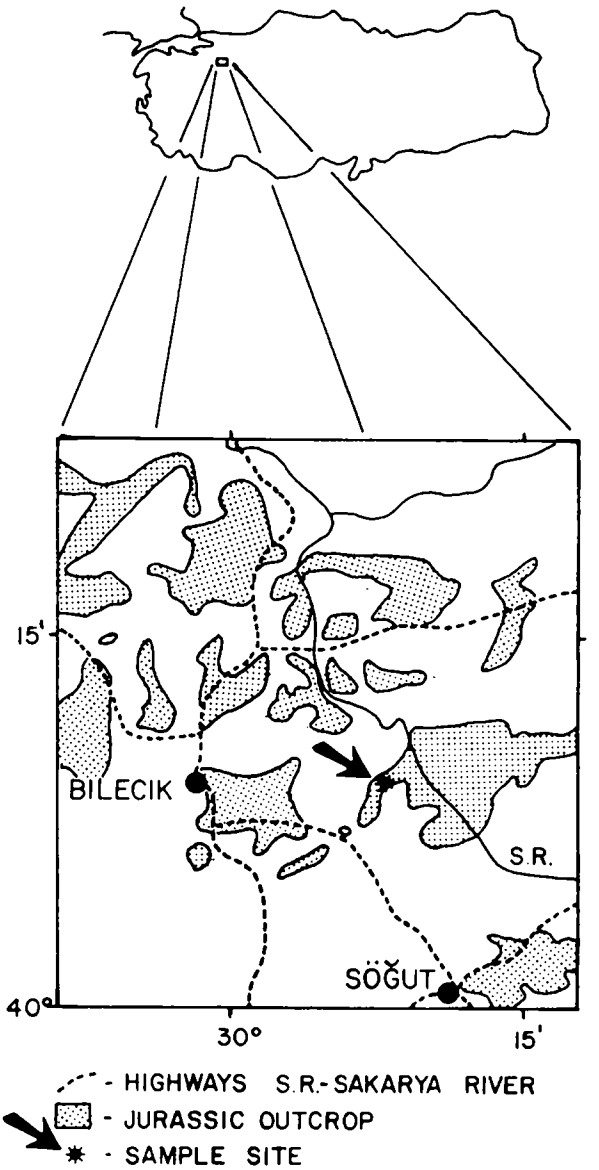


Fig. 3. Map of Bilecik area.

near 5×10^{-5} and 5×10^{-4} A/m. Only one of the samples had an intensity below the noise level. A stereonet of the NRM directions (Fig. 4b) shows a broad concentration of points in the northeast quadrant. Correction for the dip of the beds (Fig. 4c) produces much steeper inclinations. In neither case do the points concentrate around the present pole position but are distributed between the pre-

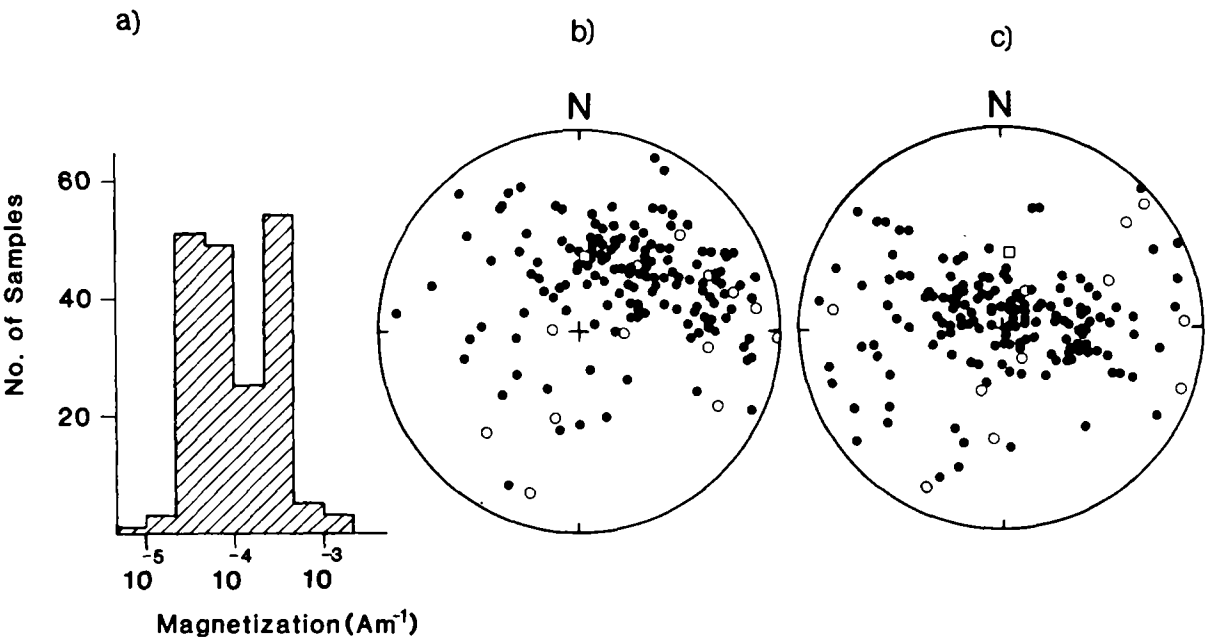


Fig. 4. (a) Histogram of NRM intensity of samples of Bilecik Limestone. Stereographic plot of NRM direction before (b) and after (c) bedding dip correction. ● = lower hemisphere, ○ = upper hemisphere, □ = present field direction.

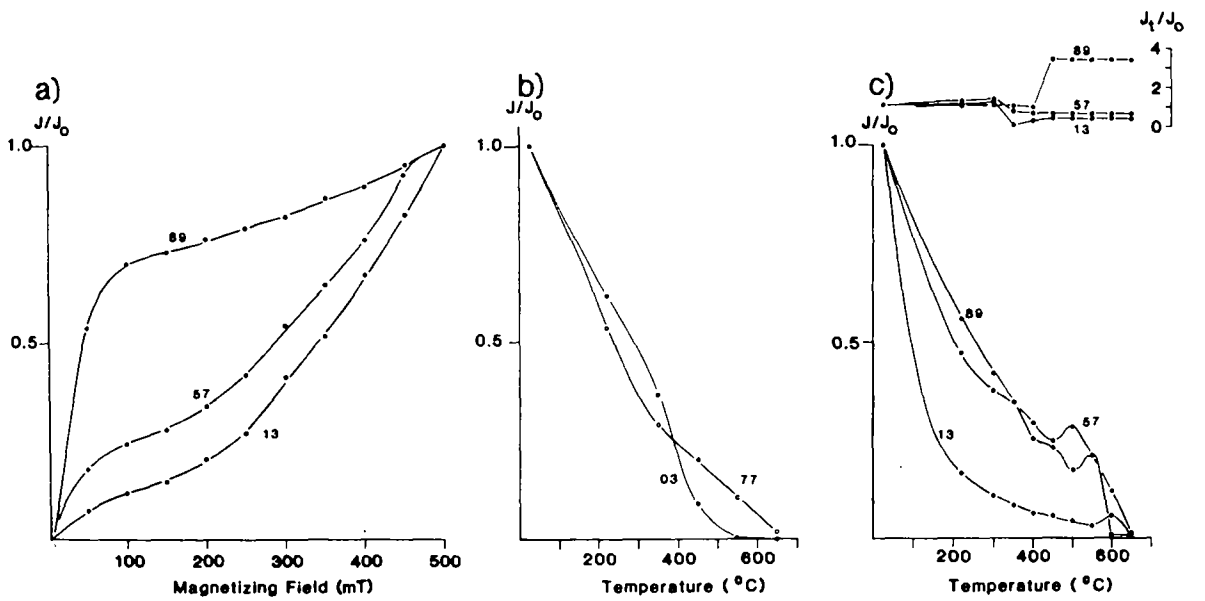


Fig. 5. (a) IRM acquired in d.c. magnetizing fields. (b) Stepwise thermal demagnetization of IRM acquired during initial exposure to a 500-mT field. (c) Stepwise thermal demagnetization of IRM acquired in a 500-mT field. Samples were exposed to a 500-mT field after each heating step.

sent field direction and an approximately east-west direction.

In order to identify the magnetic minerals present, isothermal remanent magnetization (IRM) studies of several samples were carried out. The samples were exposed to magnetizing fields from 50 to 500 mT in steps of 50 mT (Fig. 5a). Each displayed a similar behavior: rapidly acquiring an IRM in fields of 50–200 mT. This was followed by a short plateau in the magnetization curve prior to acquiring a further IRM, this time more slowly, up to 500 mT. None of the samples studied is saturated at 500 mT. The initial IRM is believed to be due to magnetite whose maximum coercivity is 300 mT. The IRM acquired above 300 mT is thought to be due to either goethite or hematite both of which have saturation fields well above 500 mT. In order to further enhance the identification of the magnetic minerals present, stepwise thermal demagnetization of the IRM was carried out in two ways. In the first method the samples were exposed to a 500-mT field only once and stepwise thermally demagnetized at 220–300°C and thereafter at 50°C intervals to 700°C. In the second method after each demagnetization step the samples were once again exposed to a 500-mT field and their IRM measured. This latter technique is used to monitor any mineralogical and/or chemical changes which may take place during heating.

Fig. 5b shows the results of the first method for two samples. The IRM of sample 03 declines with temperature to approximately 550°C above which the magnetization is less than 0.1% of the initial IRM. This indicates the presence of magnetite. In sample 77 there is a slower decline above 400°C which continues to 650° indicating the presence of hematite. Fig. 5c shows the results of the second method. The magnetization of sample 13 decreases rapidly at first then more slowly above 220°C. This behavior together with that shown in Fig. 5a shows that a substantial amount of goethite is present in this sample. With the exception of sample 13, the samples all display a regular decline in IRM to 400°C above which many show a minor increase or change in slope before declining again to 650°C. This increase is attributed to the presence of magnetite. The IRM acquired by the samples in a 500-mT field after each temperature step is also shown in Fig. 5c. As can be seen, the samples fall into two groups, those which experience an appreciable increase in their magnetization between 350 and 450°C and those which display a notable decrease over the same temperature range. Most show only a relatively minor increase between 25 and 250°C. A detailed study of the behavior of goethite during heating has been carried out by Strangway et al. [29] who found that the hematite formed from dehydration of goethite above 350°C has two types of magnetic behavior. When the grain size is greater than 0.5 μm the hematite shows characteristic ferromagnetic behavior. In smaller grain sizes, however, a weak thermal remanent magnetization (TRM) is acquired but the material displays none of the usual ferromagnetic properties [29].

To determine their stable remanent magnetizations the samples were "cleaned" using thermal demagnetization. Alternating field demagnetization was also carried out on several samples, but was not as successful as heating. This is thought to be due to the presence of goethite in the samples. Several samples were stepwise heated to 700°C and their remanent magnetization measured. Those with larger concentrations of goethite decreased in intensity rapidly (sample 75, Fig. 6a) to 220°C but above this remained stable declining only slowly to 600°C. The magnetization can be seen to be due to a single component above 220°C (Fig. 6b). The intensity of other samples decreased more slowly with temperature (Fig. 6a) some samples showing a minor increase to 220°C above which these also decreased. Although several samples displayed a change in slope between 550 and 600°C many retained a measurable magnetization to 650°C. This agrees well with the IRM studies suggesting that both hematite and magnetite are present. Vector plots of the magnetization show no appreciable change in the direction of magnetization between 550 and 650°C indicating that the magnetite and hematite components carry similar directions of magnetization. This strongly suggests that the remanence was acquired at, or shortly after, deposition.

On the basis of the vector plots of pilot samples (Fig. 6b) two components of magnetization are observed: a low-temperature component stable be-

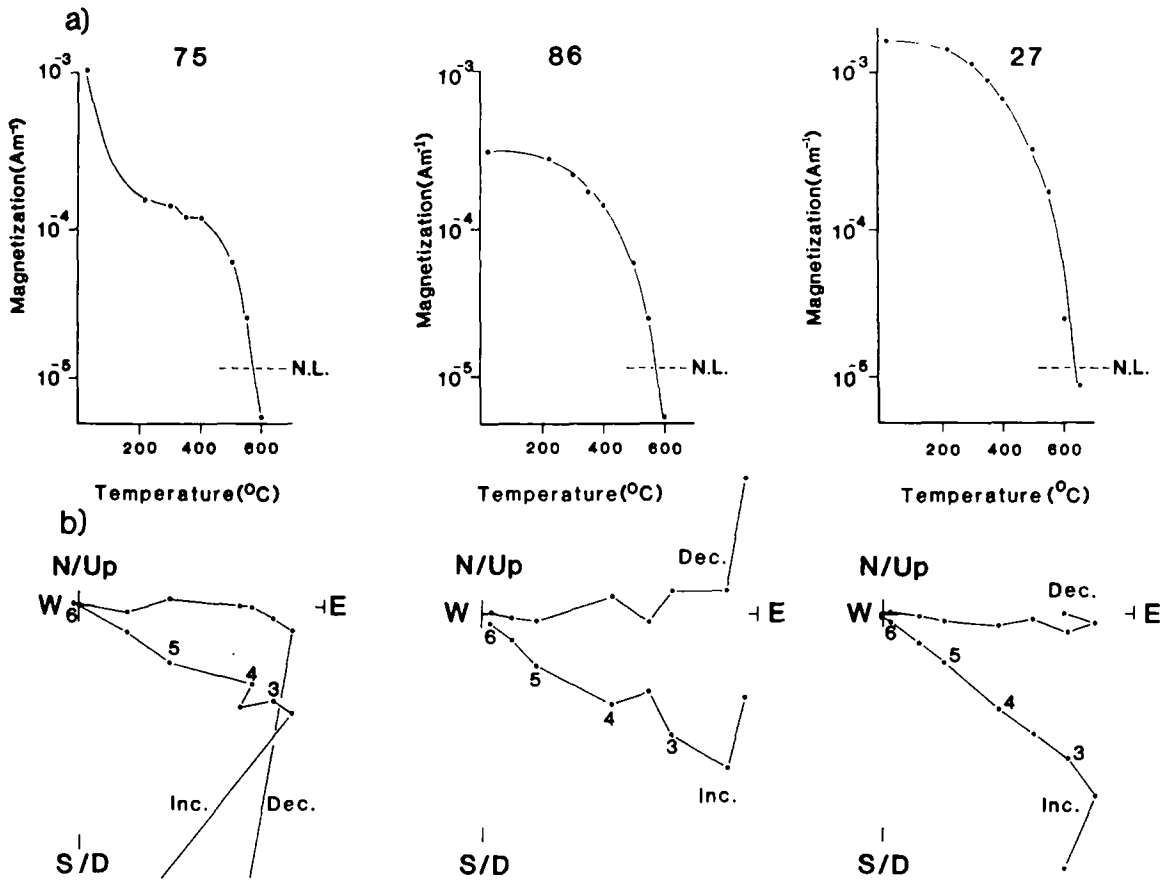


Fig. 6. (a) Stepwise thermal demagnetization of NRM for samples 27, 75 and 86, *N.L* = noise level of magnetometer. (b) Vector plots of declination and inclination during thermal demagnetization. Numbers at demagnetization steps in 100°C increments.

low 220°C and a higher-temperature component which remains stable to 550–600°C. The low-temperature component is due to the secondary mineral goethite and does not therefore represent a primary magnetism. The primary magnetization is believed to be carried by the magnetite/hematite component, and consequently, all samples were heated to 220°C and their magnetization measured.

Magnetic results. Several samples were rejected on the basis of their intensity and/or consistency of magnetization directions. The samples used were those that had less than ±5° error in the direction of the magnetization vector based on the uncertainty due to measurement errors and field corrections. The resulting 134 samples were used to obtain a paleomagnetic pole position. The sample directions shown in Fig. 7 yield a mean pole position of 18.8°N, 91.8°E ($\alpha_{95} = 7.7°$; $N = 134$). The mean declination of 92.8° indicates the section has undergone an approximately 90° clockwise rotation since the Late Jurassic with respect to the geographic axis. Paleomagnetic pole positions for Europe in the Late Jurassic [30] indicate that the section has experienced a 60–70° clockwise rotation with respect to this stable block. The mean inclination (52.7°) is shallower than expected for its present geographic location suggesting that the section may have moved slightly northward since the remanence was acquired.

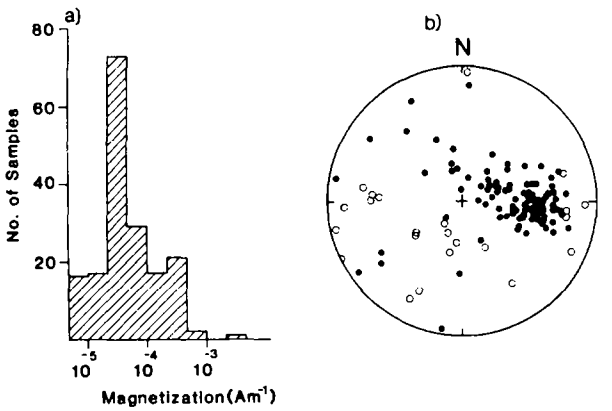


Fig. 7. (a) Histogram of remanent magnetizations after thermal demagnetization to 220°C. (b) Stereographic plot of magnetization directions (corrected for dip of the bed) after thermal demagnetization to 220°C. Symbols as in Fig. 4.

4. Discussion

A number of models have been proposed to explain the complex geology of the Alpine-Mediterranean area: ([1,31–33] and others). In this paper, the model proposed by Şengör [34], Şengör et al. [35] and Şengör and Yilmaz [36] is adopted for the sake of simplicity and because the model is weighted heavily by studies of Turkish geology.

According to this model the Neo-Tethys was a major east-west ocean that separated Laurasia and Gondwanaland. The northern margin of Gondwanaland was marked by the African and Arabian promontories, features which have been recognized for a long time [37] and which have been the subject of much debate (see Channell et al. [38] for a summary of the problem). Şengör [34] and Şengör and Yilmaz [36] suggest that the Neo-Tethys split westward into two major branches. A northern branch, the Vardar Ocean, and a southern branch that is still extant in the form of the Eastern Mediterranean Sea (Fig. 8). These two ocean branches, in this particular model, define the basic geometry and extent of the Apulian peninsula, the eastern most part of which forms much of today's central Anatolia. The central and western part of this peninsula appear to have been extremely complex as suggested by Burchfiel [39].

In placing the results of our work in the context of this model, two major problems arise. These are related to the size of the block and cause and

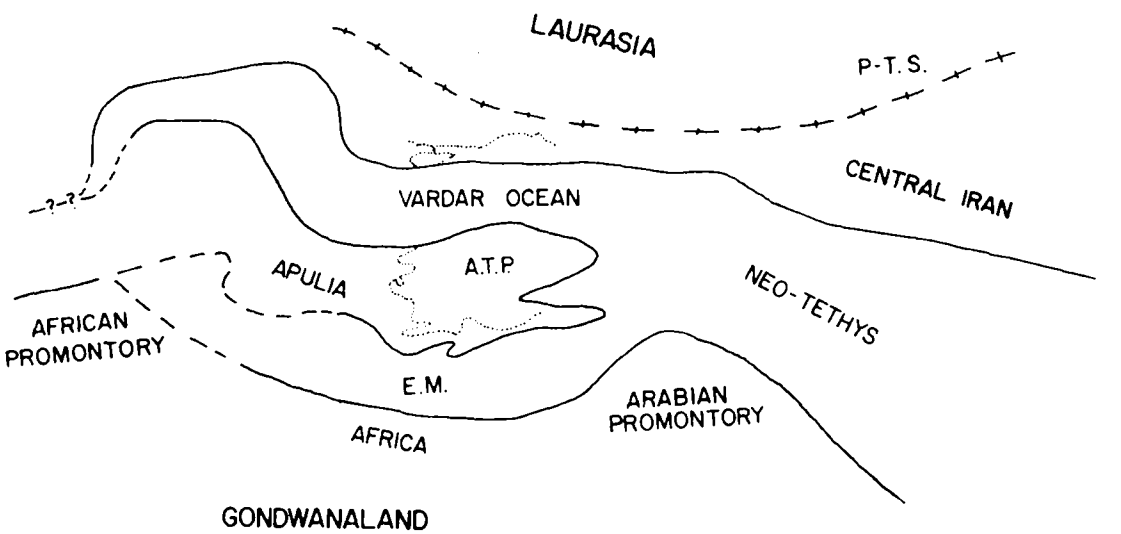


Fig. 8. Late Jurassic reconstruction of the western portion of the Neo-Tethys showing simplified geometric relationships between Gondwanaland, Apulia, and Laurasia. Oceanic widths shown are entirely schematic. *A.T.P* = Anatolide-Tauride Platform; *E.M* = Eastern Mediterranean; *P-T.S*-Paleo-Tethyan Suture. Dotted lines indicate the present geographic outline of western Anatolia for reference purposes only. Modified in part after A.M.C. Şengör (personal communication, 1979) and in part after Şengör and Yilmaz [36].

timing of its rotation. Arguments concerning the rotation of a "Turkish plate," in the Cretaceous such as suggested by Orbay and Bayburdi [21] are not valid since Turkey as we know it today probably did not exist until Miocene-Pliocene time [2]. The Bilecik Limestone was deposited in the Vardar Ocean which is believed to have had a roughly east-west orientation. The overall orientation of associated facies belts should therefore trend east-west also. In this setting, the northern margin of Gondwanaland (northern Apulia in this region) would have had its platform/shelf facies to the south and its slope/basin facies to the north. The reverse sequence would be expected on the southern margin of the Laurasia. The Jurassic facies belts of northwestern Anatolia are in fact oriented north-south with the platform/shelf facies to the west and the slope/basin facies to the east (Fig. 2). If the orientation of the facies belts has been achieved by a 90° clockwise rotation as suggested by the paleomagnetic results, the size of the rotated block would be fairly large (> 100 km $\times$ 30 km based upon continuity of facies belts and present outcrop patterns). Assuming the facies belts have been rotated through 90°, the basin geometry requires that the Bilecik Limestone formed on the northern margin of Apulia or on the northern margin of the Sakarya continent. The diagrams in Şengör and Yilmaz [36, particularly figs. 1, 6C and 6D] would suggest that this latter choice is more likely.

The rotation is probably linked to either (a) the strike-slip deformation along the North Anatolian Transform Fault initiated during the Miocene, or (b) the Paleocene to Early Eocene deformation of the Pontides associated with the collision of the Anatolide/Tauride platform with Laurasia.

The amount of motion along the North Anatolian Transform Fault is relatively small, being characterized by relatively minor offsets of 50–100 km [40]. In his paleomagnetic study of Late Cretaceous and Eocene samples from the eastern Pontides, Van der Voo [20] demonstrates that a counter-clockwise rotation can be detected in samples of the same age on both sides of the fault.

The Jurassic facies belts of northwestern Anatolia cross a major branch of the North Anatolian Fault with no significant offset [28]. The strike-slip motion of the fault therefore does not appear to significantly influence units located in its vicinity. Assuming that the strike-slip motion along the North Anatolian Fault is not responsible for the observed block rotation, it appears that the rotation has been produced by the Early Tertiary collision of Laurasia and Gondwanaland.

The above interpretation is based upon the paleomagnetic results from a single site and the existence of north-south-trending facies belts. An alternative explanation of the results is that a relatively small block has undergone a 90° clockwise rotation and that the facies belts are due to a north-south embayment of the Vardar Ocean (although this combination appears somewhat fortuitous). Further work is currently in progress to determine the size of the block involved in the rotation and also to better document the timing of the rotation by sampling units of various ages.

5. Conclusions

The paleomagnetic data gathered in this study indicate that the sample site examined has undergone a ca. 90° clockwise rotation since Middle to Late Jurassic time. Also, from the calculated pole position it appears that this site may have moved slightly northward in the same time period. Ancilliary geologic evidence, particularly the orientation of facies belts, when combined with the paleomagnetic results suggests that the Bilecik Limestone was deposited on the northern margin of the Sakarya continent, and that the clockwise rotation of this site is more likely to have been related to the collisional event that resulted in the formation of the Pontides than to the strike-slip motion associated with the North Anatolian Transform Fault.

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